

SIT723 Research Techniques & Applications

Task 3.4D Research Methodology

Introduction and Research Context

Large language models (LLMs) have shown strong performance in a variety of natural language processing tasks, such as biomedical named entity recognition, clinical question answering, and medical report generation. However, high benchmark accuracy does not always mean that a model gives consistent semantic interpretations. When a common clinical meaning is encoded in different but semantically equivalent expressions a form of unreliability that is not identified by existing evaluation systems a model can produce varied conceptual outputs.

This research addresses that gap by proposing and empirically validating a UMLS-grounded semantic entropy framework for measuring the stability of semantic interpretation in clinical NLP. Semantic entropy, introduced by Kuhn et al. (2023) and validated by Farquhar et al. (2024) for hallucination detection, measures uncertainty at the level of meaning rather than token probability. The key innovation of the present research is replacing the approximate NLI-based output clustering used in prior work with a formal UMLS Concept Unique Identifier (CUI) equivalence criterion, enabling ontology-grounded stability measurement in the biomedical domain where formal concept systems exist.

The research is grounded in four research gaps identified in the critical literature review (Task 2.3D): (1) overreliance on accuracy-based evaluation that can hide semantic instability under meaning-preserving variation; (2) absence of a UMLS-grounded semantic entropy framework for clinical NLP evaluation; (3) unknown effect of domain-specific biomedical pretraining on semantic stability under controlled perturbation; and (4) lack of a structured, ontology-grounded metric for evaluating the illusion of understanding in clinical language settings.

Research Questions

The following three research questions are expanded from Task 3.1P. Each question is directly motivated by one or more of the four research gaps identified in the literature review and is operationalised through pre-specified statistical tests, pre-registered on OSF before any data collection.

2.1 RQ1: Characterisation of Semantic Instability

Which linguistic properties of meaning-preserving perturbations perturbation type, magnitude of lexical change, and grammatical category of modified tokens are the strongest independent predictors of semantic entropy in clinical LLM outputs, and do these predictors interact?

This question addresses Gap 2 (Task 2.3D): semantic entropy has not been fully grounded in biomedical concept systems such as UMLS, and no research has systematically examined which linguistic properties of meaning-preserving perturbations drive instability in clinical language settings. Prior semantic entropy work (Kuhn et al., 2023; Farquhar et al., 2024) treats entropy as a single aggregate measure without decomposing it by linguistic dimension. This question fills that gap by using UMLS CUI clustering as the formal equivalence criterion and attributing entropy to specific input properties through a pre-registered mixed-effects regression.

Sub-questions

1. How does semantic entropy vary across perturbation types (back-translation, controlled paraphrase, synonym substitution, and syntactic reordering) when all other linguistic factors are held constant?
2. Does the magnitude of lexical change, measured as normalised Levenshtein edit distance, independently predict semantic entropy after controlling for perturbation type and grammatical category?
3. Do different grammatical categories (nouns, verbs, modifiers, function words) contribute differently to semantic entropy, and does a perturbation-type by grammatical-category interaction exist?

Statistical operationalisation

Primary analysis: Mixed-effects regression model.

Model specification: $\hat{H}(x_0) \sim \text{PerturbationType} \times \text{LinguisticCategory} + \text{Magnitude} + \text{Accuracy} + (1|\text{Instance}) + (1|\text{Model})$

Estimation: Restricted Maximum Likelihood (REML). Pre-specified minimum effect size: $f^2 \geq 0.04$ for fixed effects of interest at BH-FDR $q = 0.05$. Marginal and conditional R^2 reported. Fixed-effect 95% confidence intervals via parametric bootstrap (1,000 iterations).

Expected deliverables

- Semantic entropy scores broken down by perturbation type, lexical change magnitude, and grammatical category of modified tokens.
- Mixed-effects regression results showing independent effect sizes (f^2) for each linguistic predictor with 95% bootstrap confidence intervals.
- Test of the perturbation-type by grammatical-category interaction term and its practical significance.
- Identification of which linguistic factor most strongly drives semantic instability in clinical concept normalisation.

2.2 RQ2: Accuracy-Stability Dissociation

Does semantic entropy detect systematic interpretation instability in high-accuracy LLM outputs that traditional evaluation metrics fail to reveal, and does this accuracy stability dissociation hold across both encoder-only and generative model architectures?

Motivation and gaps addressed

This question addresses Gaps 1 and 4 (Task 2.3D). Gap 1 identifies the overreliance on benchmark accuracy despite evidence that accuracy can hide instability under meaning-preserving variation (Sclar et al., 2024; M. Agrawal et al., 2023; Hager et al., 2024). Gap 4 identifies the absence of a structured, ontology-grounded metric for evaluating the illusion of understanding in clinical language settings. This question tests whether a model performing well on accuracy can simultaneously exhibit elevated semantic entropy $\hat{H}(x_0)$ in the high-accuracy quartile, demonstrating that accuracy-based evaluation systematically underestimates reliability risk.

Sub-questions

1. Can a model achieve high task accuracy while still producing inconsistent semantic outputs across meaning-equivalent inputs specifically in the high-accuracy quartile of the output distribution?
2. Does the accuracy-stability dissociation appear in both encoder-only models (CUI-ranking entropy) and generative models (text-generation entropy), or is it architecture-specific?
3. How often do apparently correct outputs remain semantically unstable when measured through UMLS-grounded semantic entropy rather than surface accuracy alone?

Statistical operationalisation

Primary analysis: One-tailed Wilcoxon signed-rank test comparing $\hat{H}(x_0)$ in top-accuracy quartile vs. bottom-accuracy quartile per model. Pre-specified threshold: rank-biserial $r \geq 0.30$. Global dissociation test: Spearman rank correlation between per-instance accuracy and $\hat{H}(x_0)$, pooled via Fisher's z-transformation; pre-specified threshold $|\rho| < 0.30$.

Expected deliverables

- Comparison of task accuracy and normalised semantic entropy $\hat{H}(x_0)$ across the same outputs, per model.
- Statistical evidence of whether high-accuracy instances show significantly elevated semantic entropy (Wilcoxon signed-rank, one-tailed, rank-biserial $r \geq 0.30$).
- Separate analysis for encoder-only (CUI-ranking entropy) and generative (text-generation entropy) architectures.

- Results demonstrating whether semantic entropy provides reliability information that accuracy metrics cannot detect.

2.3 RQ3: Domain Adaptation and Semantic Stability

Does domain-specific biomedical pretraining reduce semantic entropy relative to general-purpose models at equivalent parameter scale, and does this stability advantage generalise from medical concept normalisation to biomedical and general-domain question answering?

Motivation and gap addressed

This question addresses Gap 3 (Task 2.3D): biomedical LLMs have not been systematically compared in terms of semantic stability under controlled meaning-preserving perturbation. While the biomedical NLP literature establishes that domain-adapted models such as BioBERT (J. Lee et al., 2020), PubMedBERT (Y. Gu et al., 2021), and BioMistral (Y. Labrak et al., 2024) outperform general-purpose models on task accuracy, it is unknown whether any accuracy advantage extends to semantic consistency. The within-scale comparison design isolates domain-specific pretraining from the confound of model scale.

Sub-questions

1. Do biomedical models show lower semantic entropy than general-purpose models of equivalent scale in within-scale comparisons: BioBERT vs. BERT-base (both 110M parameters) and BioMistral-7B vs. FLAN-T5-XXL (large-scale tier)?
2. Does any stability advantage observed on MedMentions (medical concept normalisation) persist on BioASQ (biomedical QA) and attenuate on SQuAD 2.0 (general-domain QA), indicating domain-specific rather than fully general stability benefits?
3. Does domain adaptation improve the consistency of semantic interpretation under meaning-preserving variation, or does it only improve benchmark accuracy?

Statistical operationalisation

Primary analysis: One-tailed Mann-Whitney U tests in two pre-specified within-scale pairs. Pair 1: BioBERT vs. BERT-base (both 110M). Pair 2: BioMistral-7B vs. FLAN-T5-XXL (large-scale). Pre-specified threshold: rank-biserial $r \geq 0.30$ in each pair. Generalisation condition: attenuation of advantage on SQuAD 2.0 relative to MedMentions, tested by comparing effect sizes across datasets. BH-FDR correction at $q = 0.05$ across all confirmatory tests.

Expected deliverables

- Within-scale semantic entropy comparisons: BioBERT vs. BERT-base (110M) and BioMistral-7B vs. FLAN-T5-XXL (large-scale), with effect sizes and 95% bootstrap confidence intervals.
- Semantic entropy results across all three datasets (MedMentions, BioASQ, SQuAD 2.0) enabling a domain-continuum analysis.
- Statistical evidence of whether domain adaptation improves semantic stability as well as task accuracy.
- Discussion of whether any stability advantage is domain-specific or generalises across task formats, with implications for model selection in clinical NLP.

3. Research Methods

3.1 Overview of Research Methods Considered

The following table provides a succinct overview of the main research methods relevant to the research domain, with their strengths, weaknesses, and suitability for the present research questions.

Method	Description	Strengths	Weaknesses	Suitability for this research
Semantic entropy (UMLS-grounded CUI clustering)	Computes Shannon entropy over meaning-level output clusters defined by UMLS CUI equivalence rather than token probability.	• Formal ontological grounding. • Model-agnostic. • No fine-tuning required. • Directly addresses meaning-level stability.	• Requires UMLS entity linker (noise source). • Only applicable to domains with formal concept systems.	PRIMARY METHOD. Directly operationalises all three RQs. The UMLS extension is the research's main contribution.
Accuracy-based benchmarking	Evaluates model performance against ground-truth labels using standard classification metrics.	• Well-established. • Easy to compare across studies. • Interpretable to practitioners.	• Does not capture stability under variation. • Rewards correct-once behaviour.	Used as BASELINE COMPARISON only. RQ2 specifically tests its limitations.
SelfCheckGPT (consistency-based detection)	Samples multiple outputs and checks consistency using NLI, n-gram, or BERTScore signals without formal clustering.	• Zero-resource. • No clustering required. • Multiple signal types.	• No formal equivalence criterion. • Cannot distinguish CUI-level meaning change.	Used as COMPARISON BASELINE to demonstrate the advantage of UMLS-grounded entropy over informal consistency metrics.
Meaning-preserving perturbation testing	Generates semantically equivalent input variants through back-translation, paraphrase, synonym substitution, and syntactic reordering, then measures output stability.	• Controlled exposure of instability. • Isolates linguistic factors. • Supports causal decomposition.	• Perturbation quality depends on automated pipeline. • May not capture all meaning-	CORE METHODOLOGY for generating the input variation that entropy is measured over. Supports RQ1 decomposition.

			change dimensions.	
Calibration analysis (ECE, temperature scaling)	Measures the alignment between model-reported confidence and empirical accuracy across probability bins.	- Well-understood. - Identifies overconfidence. - Can be post-hoc applied.	- Requires output probabilities. - Does not measure semantic consistency. - Cannot detect stability under paraphrase.	BACKGROUND ONLY. Helps distinguish semantic stability from calibration. Not a primary method.
Fine-tuning / domain adaptation	Adapts a pre-trained model to a specific domain by training on in-domain labelled data.	- Can improve task performance. - Reduces generalisation gap.	- Requires labelled data. - May reduce generalisability. - Not zero-shot.	NOT used in this research. All models evaluated in zero-shot setting to maximise generalisability and isolate pretraining effect.

3.2 Selected Research Method and Justification

The research adopts a UMLS-grounded semantic entropy evaluation framework as its primary method, combined with meaning-preserving perturbation generation and a six-gate automated quality validation pipeline. This combination is selected for the following reasons.

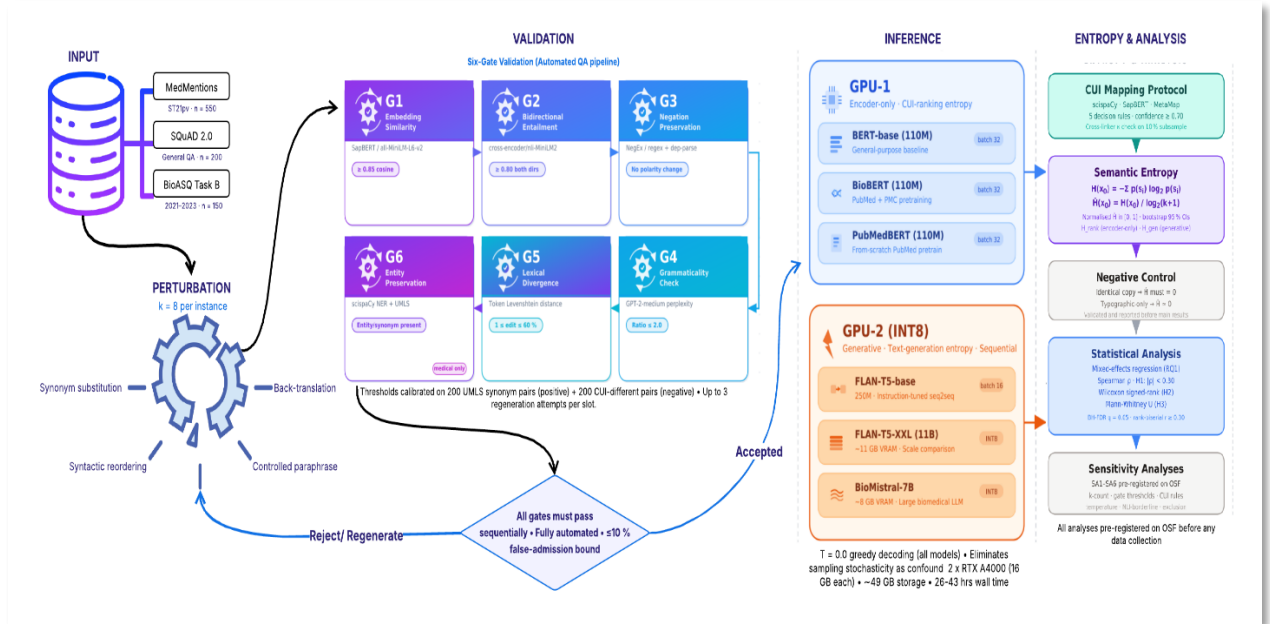


Figure 1: Semantic entropy evaluation pipeline using dataset input, perturbation generation, automated validation, model inference, CUI-based output mapping, and statistical analysis.

Why semantic entropy rather than accuracy-based benchmarking

Accuracy-based benchmarking assesses the model output in relation to whether a model provides the appropriate output on a single run of each input. It is not able to tell whether the model is always able to map semantically equivalent inputs to the same interpretation. This type of reliability is directly measured by semantic entropy (Kuhn et al., 2023; Farquhar et

al., 2024), which computes uncertainty on clusters of output, at the meaning level. This method with CUI equivalence instead of NLI-based approximate clustering as UMLS extension offers a formal ontological representation impossible in general-domain contexts, and thus the clinical NLP domain is a well-suited methodologically advantaged context of this framework.

Why meaning-preserving perturbations rather than adversarial attacks

Adversarial attacks such as TextFooler, HotFlip can frequently disobey semantic meaning to achieve prediction flips. In the current study, the strictly meaning-preserving variation is necessary to ascribe entropy to input rephrasing, as opposed to the change of meaning. The four perturbation techniques (back-translation, controlled paraphrase, synonym substitution, and syntactic reordering) are chosen since they address complementary linguistic dimensions lexical, syntactic, and translational and are all validated by the six-gate quality pipeline. The review of G. G. Şahin (2022) confirms that no one method of augmentation is superior to the rest in all tasks, which is why a multi-method approach is justified.

Why zero-shot evaluation rather than fine-tuning

All models are tested in zero-shot or structured-prompt models without task-specific fine-tuning. Three reasons have motivated this choice: (1) zero-shot evaluation maximises the generalisability of results in deployment conditions where no fine-tuning exists; (2) it separates the effect of domain-specific pretraining (the variable of interest in RQ3) and the effect of fine-tuning; and (3) it can compare biomedical and general-purpose models of the same capacity in a clean within-scale way.

Why mixed-effects regression for RQ1

The mixed-effects regression model is chosen to answer RQ1 since it appropriately models the hierarchical nature of the data multiple perturbations per instance, multiple instances per model using random intercepts, and estimates fixed effects of the linguistic predictors of interest. It is the correct structure of the entropy data that is a repeated-measures structure. The model is pre-registered on OSF prior to any data collection, which guarantees confirmatory, but not exploratory framing.

Why non-parametric tests for RQ2 and RQ3

Wilcoxon signed-rank (RQ2) and Mann-Whitney U (RQ3) are chosen since normalised semantic entropy $\hat{H}(x_0) \in [0,1]$ is bounded and does not necessarily follow a normal distribution, especially in the tails. Non-parametric tests have no distributional assumptions and are resistant to the skew that is likely to occur in entropy distributions where many cases might be concentrated at one end (stable) or at the other end (maximally unstable). Pre-specifying effect sizes (rank-biserial r) is done to guarantee that practical significance is identified as opposed to statistical significance.

4. Research Plan with Timelines

The research is organised into five phases spanning the thesis project period. Each phase has defined inputs, outputs, and dependencies. The timeline is designed to be realistic given the available hardware (2× NVIDIA RTX A4000, 16 GB VRAM each) and estimated computational requirements (26-43 hours wall time for all model inference).

Phase	Name	Key Activities	Outputs	Dependencies and Risks
1	Pre-registration and Setup	- OSF pre-registration of H1, H2, H3, SA1-SA6. - Environment setup and dependency pinning. - Threshold calibration on pilot pairs.	- Pre-registration identifier. - Docker container image. - Calibrated six-gate thresholds.	- Must be completed before any data collection. - Risk: OSF submission delay mitigated by drafting registration document in parallel with this task.
2	Dataset Preparation	- Stratified sampling of all three datasets. - scispaCy UMLS linker validation. - Pilot set reservation and exclusion check.	- Three stratified sample files. - Inclusion/exclusion log. - Pilot QA report.	- Depends on Phase 1 completion. - Risk: Higher-than-expected scispaCy rejection rate mitigated by 10% buffer sample.
3	Perturbation Generation and Validation	- Generate k=8 perturbations per instance via 4 methods. - Apply six-gate quality pipeline. - Log rejection rates per gate per method.	- Validated perturbation dataset (~15 GB). - Gate rejection rate report. - Perturbation annotation file (type, magnitude, POS category).	- Depends on Phase 2. Estimated 8-12 hours compute. - Risk: High rejection rate in G6 (entity preservation) for rare UMLS concepts up to 3 regeneration attempts per slot.
4	Model Inference and Entropy Computation	- Run all 5 models on 9 inputs per instance. - Apply CUI mapping protocol (5 rules). - Compute $\hat{H}(x_0)$ for all instances and models. - Negative control validation. - MetaMap cross-linker reliability check (10% subsample).	- Entropy dataset (n×5 models). - CUI mapping log with rule applied. - Cross-linker reliability report (kappa target ≥ 0.75). - Negative control results.	- Depends on Phase 3. Estimated 26-43 hours wall time. - Risk: VRAM overflow for FLAN-T5-XXL mitigated by INT8 quantisation and sequential GPU-2 allocation. - Risk: Cross-linker kappa < 0.75 triggers extended MetaMap reconciliation protocol.
5	Statistical Analysis and Write-Up	- Run H1 (mixed-effects regression), H2 (Wilcoxon), H3 (Mann-Whitney U). - BH-FDR correction across all tests. - Run SA1-SA6 sensitivity analyses. - Draft Results chapter. - Release code and data publicly.	- Confirmed or rejected H1, H2, H3 with effect sizes and CIs. - Sensitivity analysis report. - Results chapter draft. - Public GitHub repository and data release.	- Depends on Phase 4. - Risk: Non-normal entropy residuals in H1 pre-specified fallback to permutation test documented in pre-registration. - Risk: Insufficient n after exclusions documented as scope limitation.

4.2 Detailed Milestone Timeline

Milestone	Phase	Deliverable and Deadline	SMART Criteria
M1: OSF Pre-registration	1	- Complete OSF pre-registration of all H1/H2/H3 confirmatory tests and SA1-SA6. - Deadline: Before any data collection begins.	- Specific: Pre-registration covers all three hypotheses and six sensitivity analyses. - Measurable: OSF registration identifier obtained and recorded. - Achievable: Form drafted during Task 3.3C; submission takes 1-2 hours. - Relevant: Required for confirmatory framing. - Time-bound: Before Phase 2 begins.
M2: Dataset Preparation	2	- Stratified samples extracted: MedMentions (n=550+55 pilot), SQuAD 2.0 (n=200+20), BioASQ (n=150+15). - scispaCy validation passed. Inclusion log complete. - Deadline: 1 week after M1.	- Specific: Three dataset files with defined stratification. - Measurable: Exact n confirmed; pilot instances flagged and excluded from main analysis. - Achievable: Datasets publicly available; scispaCy runs on CPU. - Relevant: Data quality is the foundation of all subsequent analysis. - Time-bound: 1 week after M1.
M3: Threshold Calibration	2-3	- Six-gate thresholds calibrated on 200 positive (UMLS synonym pairs) and 200 negatives (different-CUI pairs). - Calibration results documented in supplementary materials. - Deadline: Before full perturbation generation.	- Specific: 400 calibration pairs, six thresholds verified. - Measurable: False-admission rate $\leq 10\%$ on negative pairs. - Achievable: Proxy pairs drawn from UMLS Metathesaurus directly. - Relevant: Gate integrity is prerequisite for perturbation quality. - Time-bound: Concurrent with M2.
M4: Perturbation Generation	3	- k=8 validated perturbations generated for all instances across three datasets. - Perturbation annotation complete (type, magnitude, POS category). - Deadline: 2 weeks after M3.	- Specific: 8 perturbations per instance, 4 methods, 6 gates. - Measurable: Rejection rate logged per gate; overall acceptance rate $\geq 90\%$. - Achievable: All perturbation models run on CPU/GPU-1. - Relevant: Perturbation dataset is the core experimental input. - Time-bound: 2 weeks; ~8-12 hours compute.
M5: Negative Control Validation	4	- Identical-copy and typographic-only control results verified for all 5 models. - Any model with $\hat{H} > 0.10$ on identical copy excluded with documentation.	- Specific: 60 instances with two control variants per instance. - Measurable: $\hat{H} = 0$ for identical copies; $\hat{H} \approx 0$ for typographic variants. - Achievable: Controls generated programmatically.

		Deadline: Before main inference run.	- Relevant: Validates entropy metric before any main analysis. - Time-bound: 2-3 hours compute after perturbation dataset ready.
M6: Model Inference	4	- All 5 models process all 9 inputs per instance across all 3 datasets. - Outputs saved to SSD with full logging. - Deadline: 2 weeks after M5.	- Specific: 5 models $\times$ (n=550+200+150) $\times$ 9 inputs = ~40,500 inference calls. - Measurable: All output logs present; no missing predictions. - Achievable: GPU-1 (encoders) and GPU-2 INT8 (generative). - Relevant: Inference is the primary data collection step. - Time-bound: 26-43 hours wall time; 2-week window allows for interruptions.
M7: Entropy Computation and Cross-Linker Check	4	$\hat{H}(x_0)$ computed for all instances and models using 5-rule CUI mapping protocol. - MetaMap cross-linker kappa check on 10% subsample (target ≥ 0.75). - Deadline: 1 week after M6.	- Specific: Full entropy dataset with bootstrap CIs. - Measurable: Kappa value reported numerically. - Achievable: CUI mapping is rule-based; entropy computation is O(n). - Relevant: Entropy dataset is the input to all statistical analyses. - Time-bound: 1 week.
M8: Confirmatory Statistical Analysis	5	- H1 (mixed-effects regression), H2 (Wilcoxon), H3 (Mann-Whitney U) conducted. - BH-FDR correction applied. SA1-SA6 sensitivity analyses conducted. - Deadline: 2 weeks after M7.	- Specific: Three confirmatory tests plus six sensitivity analyses. - Measurable: Each test produces a p-value, effect size, and 95% CI. - Achievable: All analyses run in Python (statsmodels) or R (lme4, simr). - Relevant: Directly answers H1/H2/H3. - Time-bound: 2 weeks; compute is negligible after entropy dataset is ready.
M9: Results Chapter Draft	5	- Complete results chapter submitted to supervisor for review. - Tables and figures for H1-H3 tests, sensitivity analyses, negative control, and cross-linker reliability. - Deadline: 1 week after M8.	- Specific: One full draft chapter with all tables and figures. - Measurable: Supervisor feedback obtained and documented. - Achievable: Results are mechanically derivable from M8 outputs. - Relevant: Draft is required for thesis submission. - Time-bound: 1 week after M8.
M10: Public Code and Data Release	5	- GitHub repository with pinned model checkpoints, fixed seeds (42), Docker image, and full analysis pipeline. - Public data repository with perturbation datasets and entropy scores. - Deadline: Concurrent with thesis submission.	- Specific: MIT licence; Docker image enables bit-identical reproduction. - Measurable: Repository link reported in thesis; Docker image executes without errors. - Achievable: Code maintained throughout all phases. - Relevant: Required for reproducibility and open science commitments.

			Time-bound: Final thesis submission date.
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5.1 Patient Data Privacy

All three datasets used in this research (MedMentions ST21pv, SQuAD 2.0, and BioASQ Task B Phase A) are publicly available de-identified research datasets that contain no patient identifiable information. MedMentions is derived from PubMed abstracts, which are public domain. SQuAD 2.0 is released under CC BY-SA 4.0. BioASQ is available to registered participants under the challenge data agreement. No private clinical records, electronic health records, patient notes, or personally identifiable medical data are accessed, stored, or processed at any stage of this research.

5.2 Confidentiality and Informed Consent

The research is conducted in a confidential manner by using only de-identified and publicly licensed datasets. No human subjects are enlisted, surveyed, interviewed or subjected to any experimental process. The study does not require any interaction with people and does not produce any personally identifiable results. Individual participants do not need to give informed consent. The datasets in question have the right consent and licensing measures that were put in place by their original creators during the creation of the datasets. No special informed consent procedure is required, except compliance with dataset licensing agreements that are already in place.

5.3 Clinical Safety and Hallucination Risk

The outputs of LLMs tested in this study are not utilized in any other way than research measurement. No model output is used for clinical decision-making, patient diagnosis, treatment recommendation, or any form of patient care. The research is explicitly motivated by the clinical safety concern identified by P. Hager et al. (2024, Nature Medicine) and M. Agrawal et al. (2025, npj Digital MMedicine) that high accuracy benchmarks can coexist with semantically unstable outputs. By developing a metric that goes beyond accuracy, the research directly contributes to safer evaluation practices for clinical NLP systems. The negative control condition (identical-copy perturbation) validates the entropy metric before any model comparisons are conducted, ensuring that results are not artefacts of the measurement framework.

5.4 HREC Approval and Regulatory Considerations

This research does not need any Human Research Ethics Committee (HREC) approval. The study does not use human subjects, is not deceptive, no identifiable personal information was used, and no intervention or interaction with individuals was involved. All of the data will be based on publicly available, de-identified research datasets. The decision is in line with the Deakin University Human Research Ethics process of research that does not expose human subjects to risks. The licensing conditions of the dataset (CC0 in the case of MedMentions, CC BY-SA 4.0 in the case of SQuAD 2.0, BioASQ challenge agreement) are the permissions needed to use this data in academic research.

5.5 Reproducibility and Research Integrity

All confirmatory analyses (H1, H2, H3) and sensitivity analyses (SA1-SA6) are pre-registered on OSF before any data collection or model inference. The pre-registration identifier is reported in the thesis. All code is released on GitHub under MIT licence with fixed random seeds (numpy: 42; PyTorch: 42; HuggingFace Transformers: 42) and pinned model revision hashes. A Docker container image with all dependencies pinned to exact versions enables bit-identical reproduction of all results. Full supplementary data perturbation datasets, gate scores, CUI mapping logs, model outputs, and entropy scores are released publicly alongside the thesis. Any deviation from the pre-registered protocol is documented as exploratory and clearly distinguished from confirmatory findings in the Results chapter.

5.6 Responsible Use of Artificial Intelligence

All LLM inference is performed in controlled research setting that models are objects of empirical study, not deployed decision-support tools. The study does not implement any model in clinical practice. The study clearly considers the threat of LLM overconfidence that is recognized by Kadavath et al. (2022) and the illusion of evaluation reported by M. Agrawal et al. (2025). Results will be presented with the right caveats concerning the level of the research and the boundaries of generalisation. All three datasets have the English-centric nature of the research as a scope limitation that is explicitly identified as a scope condition in the research questions and in the discussion section.

6. Milestones and Sustainability

6.1 Key Milestones Summary

The ten milestones (M1-M10) defined in Section 4.2 mark the critical stages of the research. The overall structure is designed so that no milestone is contingent on an external dependency that cannot be controlled by the research team. The only external dependencies are OSF submission (M1, resolved by pre-drafting the registration) and BioASQ data access (M2, resolved by advance registration with the BioASQ organisers). All other milestones depend only on data, code, and hardware resources that are already available.

Risk	Likelihood	Impact	Contingency Plan
G6 entity preservation rejection rate > 30% for rare UMLS concepts	Medium	Medium	- Up to 3 regeneration attempts per slot already built into pipeline. - If rate exceeds 30%, extend to 5 attempts for medical synonym substitution only. - Document deviation from pre-registration.
Cross-linker kappa (MetaMap vs. scispaCy) < 0.75	Low	High	- Trigger extended reconciliation: manually review a stratified subsample of 50 disagreement cases. - Report kappa numerically regardless of threshold outcome. - If systematic bias identified, apply rule-based correction and document.

FLAN-T5-XXL VRAM overflow under INT8 quantisation	Low	High	• Fallback: FLAN-T5-XL (3B INT8, ~3.5 GB VRAM) as alternative large-scale general model. • Document model substitution as deviation from pre-registration. • Report sensitivity to model choice in SA4.
Mixed-effects regression residuals severely non-normal (H1)	Low	Medium	• Pre-registered fallback: permutation-based mixed-effects test (5,000 permutations). • Report both parametric and permutation results. • Document as sensitivity deviation.
Dataset n falls below target after quality-gate exclusions	Low	Medium	• 10% buffer sample already built into sampling strategy. • If buffer exhausted, report actual n with power re-calculation. • SA6 (exclusion impact analysis) will verify representativeness of included instances.
GPU hardware failure or extended unavailability	Low	High	• Encoder-only models (BERT-base, BioBERT, PubMedBERT) can run on CPU, extending but not preventing completion. • Google Colab Pro (A100) available as emergency cloud fallback for generative models. • Timeline extended by maximum 2 weeks.

6.3 Sustainability Measures

Open science and data accessibility

All research outputs code, datasets, model outputs, and entropy scores will be released publicly under permissive licences (MIT for code; CC BY 4.0 for data). This ensures that other researchers can reproduce, verify, and build upon the findings without requiring access to the original hardware or institutional resources. The Docker container image ensures that results are reproducible at the bit level, not just conceptually.

Dissemination strategy

The primary target venue for journal publication is the Journal of Biomedical Informatics (JBI), which is the leading venue for work at the intersection of biomedical NLP evaluation and clinical informatics. Secondary target venues include npj Digital Medicine (if the clinical safety framing is the lead contribution) and Bioinformatics (if the UMLS-grounded entropy framework is the lead contribution). The work is also suitable for submission to the Transactions of the Association for Computational Linguistics (TACL) if the computational linguistics community is the primary audience. A conference paper version is planned for submission to ACL, EMNLP, or BioNLP workshops to obtain early peer feedback before journal submission.

Long-term relevance and extension pathways

The UMLS-grounded semantic entropy framework that will be developed in this research has clear extension pathways beyond the present scope. First, the framework is directly applicable to any biomedical NLP evaluation task where UMLS concept systems are relevant, including clinical NER, medical relation extraction, and drug-disease association mining. Second, the perturbation pipeline and six-gate quality framework are reusable components

that other researchers can apply to different datasets and models. Third, if domain-specific pretraining is found to confer stability advantages (RQ3 confirmed), this finding has direct practical implications for model selection guidelines in clinical NLP – a policy-level contribution that extends beyond the academic literature.

Computational efficiency and environmental considerations

The research uses INT8 quantisation for large generative models (FLAN-T5-XXL and BioMistral-7B) to reduce VRAM requirements and energy consumption relative to full-precision inference. All encoder-only models run on GPU-1 in batches of 32, which is near-optimal for the available hardware. The estimated total wall time of 26-43 hours is modest for research of this scope. This research will be conducted on a Deakin University provided virtual machine, which serves as the primary computing environment for all experiments. The codebase and supporting scripts will be maintained in a GitHub repository to support version control, transparency, and reproducibility. Intermediate results will be cached to minimise redundant computation.

References

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